

Unified Projection MPPI: Motion and Wrench Constraints under Model Uncertainty

Abstract—Manipulation requires feasible motion and admissible interaction wrenches, yet enforcing one can disturb the other, and model-feasible commands can violate physical limits during execution. We present Unified Projection Model Predictive Path Integral control (UP-MPPI) for motion, reaction, and uncertain support-wrench constraints. An exact separation theorem characterizes the metrics that decouple motion and reaction corrections. A physical–virtual equality distinction predicts the force sensitivity of position-level retraction to geometric model offsets. An averaging-distortion identity motivates projecting the raw motion mean while averaging projected reactions. Explicit execution margins, vertex constraints, and set-membership adaptation provide conditional wrench admissibility under parameter uncertainty. Simulations of surface writing, dual-arm transport, and nonprehensile transport evaluate these mechanisms. Task-adapted safe-sampling and optimization baselines expose tradeoffs across the tasks. Writing achieves 1.58 mm tracking error and 99.3% measured force-band occupancy. In the main transport study, the nominal controller tips the object in five of five runs; robust and adaptive variants have no tips. Force–torque-only adaptation reduces second-move error from 41.2 to 30.4 mm. Separate GPU benchmarks take 4.43 ms for writing and 8.65 ms for transport at 1024 samples. The execution certificate requires bounded discrepancy; the transport data validate its period-averaged form. Project page: <https://rcilab.khu.ac.kr/up-mppi/>.

Index Terms—Model predictive control, sampling-based control, constrained manipulation, force control.

I. INTRODUCTION

A ROBOT carrying an unsecured object must respect the support wrench needed to keep that object on its tray. A robot writing on a surface must simultaneously follow a path, maintain contact, and regulate tool orientation. A pair of arms grasping one object must preserve a closed chain without generating excessive internal loading. These tasks combine geometric constraints with forces that depend on the physical interaction and the accuracy of its model.

Model Predictive Path Integral control (MPPI) evaluates many candidate trajectories in parallel and accommodates nonconvex objectives without differentiating the task cost [1]. Its sampling structure is attractive for manipulation, but a penalty on constraint violation does not itself make the selected command feasible. Projection-Retraction MPPI (PR-MPPI) [2] handles equality and inequality constraints in joint-velocity space and retracts the executed configuration onto the equality manifold. Contact-Aware Manifold Projection MPPI (CAMP-MPPI) [3] instead projects sampled commanded accelerations under kinematic and wrench constraints, using the inertia metric and constrained dynamics.

These two formulations leave a common set of questions. When can motion and wrench corrections be solved separately? Which equalities should the physical interaction enforce, and which must the controller enforce? How should the weighted

sampling update be formed once an inequality projection is nonlinear? Finally, what does a hard model constraint imply when object parameters and the executed acceleration are uncertain?

We address these questions through a common projection geometry. At a fixed state, the commanded acceleration decomposes into a component that changes the realized motion and a component that changes the constraint reaction. The corresponding transformation is an affine bijection. Its geometry yields an exact condition under which a projection onto motion and reaction inequalities separates. This condition is stronger than choosing an inertia-weighted objective: it identifies the entire class of separating metrics and the constraints that invalidate separation.

The same distinction clarifies geometric model error. A physical contact supplies a reaction, whereas a desired tool orientation does not. Retracting a position reference onto an incorrectly placed physical manifold can convert a small geometric error into a large force. We derive the local sensitivity and an anchored order-1 alternative. At order 2, a constant offset of the equality function leaves its Jacobian and the predicted reaction map unchanged; this invariance has a precise, limited scope and does not remove errors in the contact normal or inertia.

Projection also changes the sampling update. Averaging projected samples and projecting their raw average are generally different operations. We derive an exact expression for that difference and show that, on a face active at the projected raw average, it points toward the feasible side. We project the raw motion mean to avoid this displacement, while averaging admissible sample reactions to truncate normal exploration. Separation allows these choices to coexist. The required execution margin is stated explicitly in the rows. For uncertain interaction models, vertex constraints and bounded-error parameter updates preserve a conditional wrench certificate while reducing conservatism online.

The contributions are:

- 1) A necessary and sufficient metric condition for separating motion and reaction projections, together with the effects of coupled rows and underactuation (Section III-A).
- 2) A physical–virtual equality assignment, a local force-error law for position-level retraction, and an anchored alternative (Section III-B).
- 3) A projected sampling controller with an exact averaging-distortion identity, separate motion and reaction updates, explicit execution-error margins, and robust set-membership adaptation (Sections III-C and III-D).
- 4) Simulation studies with matched averaging ablations, task-adapted safe-sampling and optimization baselines,

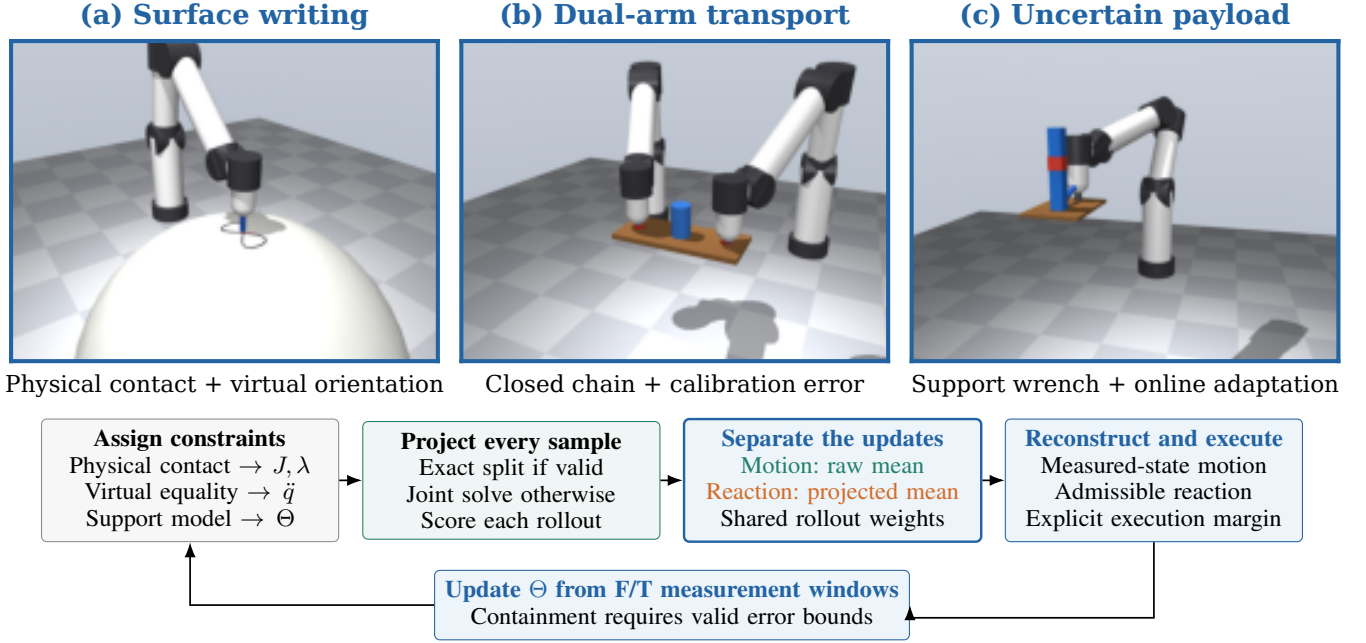


Fig. 1. UP-MPPI across three manipulation tasks (top: recorded simulation states, seed 0) and its control cycle (bottom). Physical contact supplies a reaction; virtual equalities constrain motion. Under the separation conditions, the controller combines a projected raw motion mean with averaged projected reactions. Blue identifies the proposed method throughout the figures and tables. Robust rows, explicit execution margins, and bounded-error parameter updates connect model feasibility to the conditional execution claim.

and GPU timing of separable and joint projections (Section IV).

The first-order projection and retraction mechanism is attributed to PR-MPPI, and hard acceleration-level wrench projections are attributed to CAMP-MPPI. The present contribution is the structure, interpretation, and uncertainty treatment connecting these mechanisms. The empirical studies below are new simulations and computational benchmarks; physical experiments reported in the cited papers are not counted as validation of UP-MPPI.

A. Related work

Motion and force control: Hybrid position/force control [4] and operational-space control [5] provide the classical basis for separating motion and force. Gauss’s principle expresses constrained acceleration as an inertia-metric projection [6], [7]. Hierarchical quadratic programs and task-space inverse dynamics extend instantaneous control to multiple tasks and unilateral constraints [8], [9]; control barrier functions provide inequalities for safety-critical feedback [10], [11]. We use this established geometry. The additional result is a characterization of separating metrics for inequality-constrained projections and its consequence for every sample in a predictive controller.

Constrained sampling-based control: Information-theoretic MPC motivates exponential trajectory weighting and moment updates [1]. Robust MPPI studies disturbance rejection and performance guarantees [12]. Shield-MPPI [13], composite-barrier safe sampling [14], and reachability-based DualGuard MPPI [15] combine sampling with safety mechanisms. These works establish that constraining MPPI

samples or filtering their output is not itself new. PR-MPPI [2] specializes projected sampling to constraint-manifold control, while CAMP-MPPI [3] adds hard dynamics-level wrench constraints. Our analysis asks when those projections can be decomposed, how physical and virtual equalities differ, and how uncertainty and execution error enter a wrench certificate. Section IV-C compares these mechanisms on the common transport task. Appendix A-F specifies the local adaptations and retains the extended contact-task comparisons.

Robust adaptation and object transportation: Robust MPC uses restricted constraints to account for bounded disturbances [16], [17]. Set-membership adaptive MPC maintains feasible parameter sets and updates them recursively [18], [19]; robust adaptive barrier functions similarly combine estimation and safety filtering [20]. These principles motivate the parameter-containment argument used here. Its contribution is their integration with the sampled interaction-wrench constraints and the measurements available during the manipulation task, rather than a new general set-membership theorem.

For nonprehensile transport, Heins and Schoellig study support constraints with obstacle avoidance [21] and robust planning under uncertain inertial parameters [22]. The latter provides an important precedent for enforcing balancing conditions over an uncertain parameter set. Serving Time uses reachability to certify transport of unsecured objects [23]. Zhang *et al.* combine safe exciting trajectories with bounded-error payload identification [24]. Our transport study uses measurements generated by the transport motion to update a center-of-mass interval inside a projected sampling controller. Offline robust planning and online robust NMPC provide additional comparators; their solvers, discretizations, and constraint treatment are specified in Appendix A-F.

TABLE I
RELATIONSHIP TO THE TWO PRECEDING METHODS. THE FIRST TWO COLUMNS DESCRIBE ESTABLISHED MECHANISMS; THE FINAL COLUMN STATES THE FOCUS OF THIS PAPER.

PR-MPPI [2]	CAMP-MPPI [3]	Present work
Velocity projection; executed retraction Geometric equality enforcement	Acceleration projection; constrained dynamics Contact-manifold and wrench rows	Common geometry and separating metrics Physical-virtual assignment and offset sensitivity
Projected-sample update	Projected-sample update	Separate channel averages; explicit execution margins
Known geometric model	Known interaction model	Robust parameter sets and task-time adaptation

II. PROBLEM FORMULATION

Consider a fully actuated manipulator with configuration $q \in \mathbb{R}^n$, inertia $M(q) \succ 0$, and bias force $b(q, \dot{q})$. A physical equality $c_p(q) = 0$ has Jacobian $J = \partial c_p / \partial q \in \mathbb{R}^{m \times n}$. Throughout the local analysis, J has full row rank and its smallest singular value is bounded away from zero. The commanded acceleration a is applied through inverse dynamics,

$$M\ddot{q} + b = \tau + J^\top \lambda, \quad \tau = Ma + b. \quad (1)$$

Combining (1) with $J\ddot{q} + \dot{J}\dot{q} = 0$ gives

$$\begin{aligned} \Lambda &= (JM^{-1}J^\top)^{-1}, & F &= I - M^{-1}J^\top \Lambda J, \\ f &= -M^{-1}J^\top \Lambda \dot{J}\dot{q}, \\ \ddot{q}(a) &= Fa + f, & \lambda(a) &= -\Lambda(Ja + \dot{J}\dot{q}). \end{aligned} \quad (2)$$

Equivalently, $\ddot{q}(a)$ minimizes $\frac{1}{2}\|x - a\|_M^2$ subject to the acceleration equality, where $\|z\|_W^2 = z^\top Wz$. This is the Gauss projection [6], [7]. With no physical equality, $F = I$, $f = 0$, and the reaction channel is absent.

A *virtual equality* $c_v(q) = 0$ is a desired relation for which the environment supplies no reaction, such as tool orientation or tray levelness. At order 2 it is imposed on the realized acceleration through

$$J_v \ddot{q} + \dot{J}_v \dot{q} = -K_D J_v \dot{q} - K_P c_v. \quad (3)$$

It must not be added to J in (2) unless the physical mechanism actually enforces it.

For a safe set $h(q) \geq 0$, an order-2 barrier row is

$$\nabla h^\top \ddot{q} + \dot{q}^\top \nabla^2 h \dot{q} + (k_1 + k_2)\dot{h} + k_1 k_2 h \geq 0. \quad (4)$$

At a fixed state this is affine in a [11]. Joint limits, velocity limits, virtual equalities, reaction bands, and support-wrench conditions therefore form a polyhedron $\mathcal{A}(x)$ in commanded acceleration. The sampling projection is

$$\Pi_x^W(v) = \arg \min_{a \in \mathcal{A}(x)} \frac{1}{2}\|a - v\|_W^2. \quad (5)$$

All statements about this projection require a nonempty feasible set; when some rows are relaxed, only the rows retained as hard constraints keep their guarantee.

The corresponding order-1 controller uses $u = \dot{q}$, $q^+ = q + \Delta t u$, and

$$\Pi_q^{(1)}(v) = \arg \min_u \frac{1}{2}\|u - v\|_W^2 \quad \text{s.t.} \quad J_c u = 0, \quad J_h u \geq -\Gamma \bar{h}, \quad (6)$$

with $\bar{h} = h - h_{\text{safe}}$. PR-MPPI combines this projected rollout with retraction of the executed configuration [2]. We do not rederive that controller; Section III-B analyzes how its reference interacts with a physical constraint.

For unknown interaction parameters θ , the wrench model is $w(a; \theta) = G(\theta)a + g(\theta)$ and the admissibility rows are $A(\theta)w \leq d(\theta)$. Our objective is to optimize a finite-horizon task cost while enforcing these rows for every θ in a maintained parameter set and accounting explicitly for a bounded discrepancy between model and execution. This is a per-cycle constraint objective. It does not assume that a feasible command exists at every future state.

III. UNIFIED PROJECTION MPPI

The method follows four connected steps: identify the geometry that separates motion and reaction, assign physical and virtual equalities to their appropriate models, construct the projected sampling update, and account for parameter uncertainty and execution error in the constraint rows.

A. Motion-reaction geometry

Fix the measured state and define

$$\mathcal{T} = \ker J, \quad \mathcal{N} = \text{range}(M^{-1}J^\top). \quad (7)$$

The commanded acceleration has one component in $\mathcal{T}$ that changes motion and another in $\mathcal{N}$ that changes reaction. A row $c_i^\top a \leq d_i$ is a *motion row* if $c_i^\top M^{-1}J^\top = 0$, a *reaction row* if $c_i \in \text{range}(J^\top)$, and a *coupled row* otherwise. Motion rows can be expressed using $\ddot{q}(a)$ alone; reaction rows using $\lambda(a)$ alone. Torque limits generally produce coupled rows.

Theorem 1 (Separation of inequality projections). *Let $\Phi(a) = (\ddot{q}(a), \lambda(a))$ be defined by (2).*

- 1) $\mathbb{R}^n = \mathcal{T} \oplus \mathcal{N}$ and $\mathcal{T} \perp_M \mathcal{N}$. The map Φ is an affine bijection onto $(f + \mathcal{T}) \times \mathbb{R}^m$, with

$$\Phi^{-1}(x, \lambda) = (x - f) - M^{-1}J^\top(\lambda + \Lambda \dot{J}\dot{q}). \quad (8)$$

- 2) For any two commands,

$$\|a - a'\|_M^2 = \|\ddot{q}(a) - \ddot{q}(a')\|_M^2 + \|\lambda(a) - \lambda(a')\|_{\Lambda^{-1}}^2. \quad (9)$$

- 3) Suppose the constraints contain only motion and reaction rows and define nonempty closed convex sets $\mathcal{Q} \subset f + \mathcal{T}$ and $\Omega \subset \mathbb{R}^m$. Then the M -projection of v onto $\Phi^{-1}(\mathcal{Q} \times \Omega)$ is

$$a^* = \Phi^{-1}\left(\text{proj}_{\mathcal{Q}}^M \ddot{q}(v), \text{proj}_{\Omega}^{\Lambda^{-1}} \lambda(v)\right). \quad (10)$$

- 4) A metric $W \succ 0$ separates these two projections for every input and every such pair of sets if and only if

$$\text{range}(W^{-1}J^\top) = \text{range}(M^{-1}J^\top). \quad (11)$$

Proof. For $x \in \mathcal{T}$ and $y = M^{-1}J^\top z$, $x^\top M y = (Jx)^\top z = 0$. The intersection of the two subspaces is zero because $JM^{-1}J^\top$

is nonsingular. Their dimensions sum to n . The projector F selects the tangential component; J selects the reaction component. This proves the bijection and (8).

The two components of $a - a'$ are $\Delta\ddot{q}$ and $-M^{-1}J^\top\Delta\lambda$. Their squared M -norms add, and the second norm is $\Delta\lambda^\top\Lambda^{-1}\Delta\lambda$, proving (9). The objective therefore separates on the product $\mathcal{Q} \times \Omega$, giving (10).

For a general W , separation holds if $\mathcal{T} \perp_W \mathcal{N}$. This is equivalent to $W\mathcal{N} \subset \text{range}(J^\top)$, and equality of dimensions gives (11). Conversely, if (11) fails, some $W^{-1}J^\top z$ has a nonzero tangential component. Projection onto a violated single reaction half-space with normal $J^\top z$ changes the command along $W^{-1}J^\top z$ and therefore changes its motion. Universal separation is impossible. $\square$

The familiar inertia metric is one solution, not the only one. Every $W = M + J^\top S J$, $S \succeq 0$, also satisfies (11). The Euclidean metric satisfies it only when $\text{range}(J^\top)$ is M -invariant. Thus a force correction under $W = I$ may alter motion even if the corresponding row depends only on reaction.

Corollary 1 (Feasibility of independent channels). *Under Theorem 1, $\mathcal{A}$ is nonempty exactly when both $\mathcal{Q}$ and Ω are nonempty. If Ω is nonempty, adding its reaction rows does not restrict the admissible realized accelerations $\mathcal{Q}$.*

The result follows immediately from the bijection. It provides a computational and a physical interpretation: a normal-force clamp can be solved separately from the tangential program, and any feasible reaction can accompany any feasible motion under the stated actuation and constraint assumptions.

Coupled constraints: If n_p coordinates are unactuated and S_a selects the actuated coordinates, realizable commanded accelerations belong to

$$\mathcal{U} = -M^{-1}b + M^{-1}S_a^\top \mathbb{R}^{n-n_p}. \quad (12)$$

Since Φ is a bijection, $\Phi(\mathcal{U})$ has dimension $n - n_p$ in the n -dimensional motion–reaction space. Its intersection with $\mathcal{Q} \times \Omega$ is generally not a product. A passive body held between two arms exemplifies this coupling: its equations of motion tie its acceleration to the grasp reactions. Torque limits and mixed wrench measurements can likewise connect the channels. These cases require a joint projection; independently feasible channel commands need not be jointly realizable.

By contrast, the support wrench of a box resting on a tray is a function of the tray’s realized acceleration. When the box is represented through support inequalities rather than included as a constrained passive coordinate, those inequalities are *motion rows*. The transport study below evaluates interacting motion constraints and uncertainty; it is not evidence for a speed benefit from motion–reaction separation. That benefit is tested on the writing task.

Predictive weighting: At a fixed state, a reaction-only correction leaves the rollout acceleration unchanged. This statement does not imply that changing reaction constraints leaves the final MPPI command unchanged. A reaction correction can change the rollout cost and hence the weights assigned to different motions. Theorem 1 applies to each projection and to averages with fixed weights; feasibility of the final projection does not depend on how the weights were obtained.

B. Physical and virtual equalities

A physical equality describes a relation enforced by a contact or mechanism. A virtual equality describes a control objective. Confusing them changes the force predicted by the model. In surface writing, the board supplies the normal reaction but does not supply the torque needed to orient the pen. In dual-arm transport, the grasped object enforces the relative grasp relation, while desired tray levelness remains a virtual constraint.

Proposition 1 (Sensitivity to a geometric offset). *At order 2, replacing $c_p(q)$ by $c_p(q) - \delta$ for a constant δ leaves $\ddot{q}(a)$ and $\lambda(a)$ in (2) unchanged at a fixed measured state.*

At order 1, suppose the command q^+ lies on the model manifold $c_p(q^+) = 0$ but the true equilibrium contact is $c_p(q) = \delta - K_{\text{env}}^{-1}\lambda$. Assume a local position loop of stiffness $K \succ 0$, a positive definite contact stiffness K_{env} , and a bilateral or closed unilateral contact. At static equilibrium,

$$\lambda = (\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1}\delta + O(\|\delta\|^2), \quad (13)$$

$$\Lambda_K = (JK^{-1}J^\top)^{-1}.$$

Proof. A constant offset changes neither J nor $\dot{J}$, proving the first assertion. For the second, write $q = q^+ + \Delta$. Static balance gives $K\Delta = J^\top\lambda$. Linearizing the contact relation gives $J\Delta = \delta - K_{\text{env}}^{-1}\lambda + O(\|\Delta\|^2)$. Substitution yields (13). $\square$

The force sensitivity is the series combination of loop and contact stiffness. For unilateral contact, the pressing sign of the offset produces this load; the opposite sign can cause lift-off, where the closed-contact derivation no longer applies. The invariance at order 2 concerns an additive offset of the equality function, not arbitrary geometric uncertainty. Errors that change a normal, Jacobian, or contact state remain relevant.

The stiffness must reflect the executed reference. For an interpolated order-1 reference restarted at the measured configuration every period and tracked with gains $M\omega^2$ and $2M\omega$, the period-averaged local stiffness is

$$K_{\text{eff}} = M(\omega^2/2 + 2\omega/\Delta t). \quad (14)$$

Using only the proportional gain would mispredict the offset-force slope. For a closed chain, $J = [J_1, -J_2]$ and $K = \text{diag}(K_1, K_2)$; (13) then predicts internal loading from a base-calibration offset.

Proposition 2 (Anchored physical retraction). *At order 1, retract physical rows onto $c_p(q^+) = c_p(q_0)$ through the measured state, and virtual rows onto $c_v(q^+) = 0$. To first order, the next equilibrium reaction is*

$$\lambda^+ = B(K_{\text{env}}^{-1}\lambda_0 + \delta(q^+) - \delta(q_0)), \quad B = (\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1}. \quad (15)$$

For a constant offset and a rigid environment, the offset-induced reaction vanishes. For a compliant environment with locally constant stiffnesses, it contracts by BK_{env}^{-1} .

Proof. Use $c_p(q) = c_p(q_0) + \Lambda_K^{-1}\lambda^+$ in the contact relation, and eliminate $c_p(q_0) = \delta(q_0) - K_{\text{env}}^{-1}\lambda_0$. This gives (15). The contraction matrix is similar to a positive definite matrix with eigenvalues strictly between zero and one. $\square$

Anchoring eliminates calibration-induced loading, but it does not regulate a desired contact force. Without a separate force-feedback law, repeated anchoring can remove the writing force itself. At order 2, physical rows instead enter (2) and reaction bounds, while virtual rows enter (3). Appendix A-B gives the discrete virtual-error dynamics and their gain conditions.

C. Projected sampling and execution

Given a nominal sequence $U = (u_0, \dots, u_{H-1})$, sample $V^k = U + \epsilon^k$. Exploration is Gaussian before componentwise clipping at $\pm 3\sigma_i$. At each rollout state, project v_t^k using (5), integrate the realized dynamics, and evaluate

$$S^k = \sum_{t=0}^{H-1} \ell(x_t^k, \lambda_t^k) \Delta t + \varphi(x_H^k),$$

$$w_k = \frac{e^{-(S^k - S_{\min})/\beta}}{\sum_j e^{-(S^j - S_{\min})/\beta}}. \quad (16)$$

The order-2 rollout is semi-implicit Euler: $\dot{q}_{t+1} = \dot{q}_t + \Delta t \ddot{q}_t$, $q_{t+1} = q_t + \Delta t \dot{q}_{t+1}$. Samples are scored after projection, so their costs describe the constrained motions used to construct the update.

Averaging distortion: Let $\hat{v} = \sum_k w_k v_k$ be the first raw-input average and $\bar{a} = \sum_k w_k \Pi(v_k)$ the average of the projected first inputs. Because these inputs share the measured state and the set $\mathcal{A}_0$ is convex, $\bar{a} \in \mathcal{A}_0$. The alternative $\Pi(\hat{v})$ is also feasible, but it need not equal $\bar{a}$.

Proposition 3 (Exact averaging-distortion identity). *Let Π be the W -projection onto a nonempty polyhedron, and let $\mathcal{F}$ be the face whose relative interior contains $\Pi(\hat{v})$. Write $P_{\mathcal{F}}$ for the affine W -projection onto $\text{aff } \mathcal{F}$. Let $\mathcal{D}$ contain samples whose active set differs from that at $\Pi(\hat{v})$. Then*

$$\bar{a} - \Pi(\hat{v}) = \sum_{k \in \mathcal{D}} w_k (\Pi(v_k) - P_{\mathcal{F}}(v_k)), \quad (17)$$

and, with $w(\mathcal{D}) = \sum_{k \in \mathcal{D}} w_k$,

$$\|\bar{a} - \Pi(\hat{v})\|_W \leq 2 \sqrt{w(\mathcal{D}) \sum_k w_k \|v_k - \hat{v}\|_W^2}. \quad (18)$$

For every row binding on $\mathcal{F}$, the distortion has nonpositive row residual.

Proof. The affine map satisfies $\sum_k w_k P_{\mathcal{F}}(v_k) = P_{\mathcal{F}}(\hat{v}) = \Pi(\hat{v})$. Samples with the same active set have $\Pi(v_k) = P_{\mathcal{F}}(v_k)$, giving (17). Both projections are W -nonexpansive and agree at $\hat{v}$, so each remaining difference is bounded by $2\|v_k - \hat{v}\|_W$. Cauchy–Schwarz gives (18). Each projected point satisfies a binding row, whereas its affine-face projection satisfies that row with equality, proving the sign assertion. $\square$

The identity attributes the difference to heterogeneous clipping. It vanishes when all samples share the relevant active set. Otherwise, averaging projected inputs introduces an interior displacement on the binding rows. Its consequences depend on the channel: in motion it can reduce progress toward a constrained goal, whereas in reaction it truncates exploration outside the admissible force band. Its size is not an execution-error bound. We therefore choose the two channel updates

Algorithm 1: UP-MPPI, one order-2 control cycle

Require: Measured x_0 ; nominal U ; $K, H, \Delta t$; parameter set Θ ; execution-error bounds; constraint priorities.

- 1: **for** $k = 1, \dots, K$ **in parallel do**
- 2: Sample clipped perturbations; set $x_0^k = x_0$.
- 3: **for** $t = 0, \dots, H - 1$ **do**
- 4: Form motion, virtual, reaction, and robust rows at x_t^k ; apply execution tightening.
- 5: Project $v_t^k = u_t + \epsilon_t^k$ using the separating metric when permitted by Theorem 1, otherwise a joint program.
- 6: Propagate x_{t+1}^k using the realized acceleration and accumulate the task cost.
- 7: **end for**
- 8: **end for**
- 9: Compute w_k and $\hat{v}_t = \sum_k w_k v_t^k$.
- 10: Rebuild rows at x_0 and project the raw mean: $b_0 = \Pi_{x_0}(\hat{v}_0)$.
- 11: **if** motion/reaction sets form a product and reaction averaging is enabled **then**
- 12: Average projected reactions: $\bar{\lambda}_0 = \sum_k w_k \lambda_0(a_0^k)$; form a_0^* using (20).
- 13: **else**
- 14: Set $a_0^* = b_0$.
- 15: **end if**
- 16: Execute the inverse-dynamics command.
- 17: Shift the raw means, append a zero seed, and reset nominal reactions when enabled, as described below.
- 18: Update Θ from valid measurement windows for the next cycle.

Fig. 2. The evaluated update uses a projected raw motion mean and an averaged projected reaction. Infeasible lower-priority rows may be relaxed; retained hard rows must remain feasible. A failure of the hard-row program is reported as such and is not covered by the certificate.

separately and place the required execution margin explicitly in the rows (Section III-D).

Channel-specific averaging: Let a_t^k denote the projected sample inputs, and let λ_0 denote the reaction map at the measured state. With $W = M$ and no coupled rows, the evaluated controller executes

$$a_0^* = \Phi_0^{-1}(\text{proj}_{\mathcal{Q}_0}^M \bar{q}_0(\hat{v}_0), \bar{\lambda}_0), \quad \bar{\lambda}_0 = \sum_k w_k \lambda_0(a_0^k). \quad (19)$$

The motion argument lies in $\mathcal{Q}_0$ by projection. Every first-step reaction belongs to the same convex set Ω_0 , so $\bar{\lambda}_0 \in \Omega_0$. This combines the projected raw motion mean with the mean of admissible sample reactions. It avoids motion averaging distortion without passing the full normal exploration through the final force command. For a scalar reaction interval, each sample reaction projection is a clamp.

An equivalent implementation first computes $b_0 = \Pi_{x_0}(\hat{v}_0)$ and then replaces its reaction:

$$a_0^* = b_0 + M_0^{-1} J_0^\top (\lambda_0(b_0) - \bar{\lambda}_0). \quad (20)$$

The added term is in $\mathcal{N}_0$, leaving the realized motion unchanged. If there is no constrained reaction channel, the update reduces to $a_0^* = b_0$. Coupled rows invalidate independent replacement; that case uses the joint final projection itself.

Control cycle: The final projection uses the measured state. Although projected inputs at the first rollout step share one feasible set, future steps belong to different state-dependent sets; an averaged future sequence is not automatically feasible. Replanning and the final channel construction establish only the current-cycle result.

The shifted nominal sequence is a warm start; its future reactions need not be feasible. Appendix A-E specifies the measured-state reaction reset used to initialize the next search.

If the reaction set Ω is nonempty and there are no coupled rows, relaxing motion rows does not affect the projected reaction. Theorem 1 therefore permits motion priorities while retaining reaction admissibility. With coupled rows, the hard reaction and coupled rows must instead be jointly feasible in a joint or lexicographic program. No reaction guarantee is asserted if that hard subset is infeasible.

Per-cycle guarantees: If the retained hard rows are feasible at the measured state, the final command satisfies them by projection and, when enabled, convex reaction averaging in the product set. Under uncertainty, Section III-D specifies which wrench claims transfer to execution. The finite-step barrier layer, virtual-error dynamics, and off-manifold rollout bounds in the appendices have additional smoothness, boundedness, and gain assumptions. None of these statements alone establishes recursive feasibility or continuous-time safety for an unknown low-level execution system.

Implementation details and the distinction between adaptive-temperature execution and fixed-temperature sampling analysis are given in Appendix A-E.

D. Uncertainty and execution margins

Hard constraints apply to the model supplied to the projection. An incorrect center of mass can make a model-admissible transport command destabilize the actual object. We separate uncertainty in the interaction parameters from discrepancy in execution.

Robust rows: At a fixed state, define the robust set

$$\mathcal{A}_\Theta = \{a : r_i(a, \theta) \leq 0 \text{ for all } i, \theta \in \Theta\}, \quad (21)$$

where $r_i(a, \theta) = A_i(\theta)(G(\theta)a + g(\theta)) - d_i(\theta)$ is affine in a for fixed θ .

Proposition 4 (Exact vertex reduction). *Let Θ be a box and, for every fixed a , let each residual $r_i(a, \theta)$ be convex separately in every coordinate of θ . Then*

$$\mathcal{A}_\Theta = \bigcap_{v \in \text{vert } \Theta} \mathcal{A}_{\{v\}}. \quad (22)$$

Multiaffine residuals satisfy this condition. Jointly affine residuals admit the same reduction for an arbitrary parameter polytope.

Proof. A convex function on an interval is bounded above by the larger endpoint value. Apply this fact successively to each parameter coordinate. Every residual is therefore bounded above by a vertex residual. The converse follows because each vertex belongs to Θ . $\square$

The resulting set remains a polyhedron in a . The number of vertices can grow exponentially with parameter dimension, so this construction is most useful for low-dimensional uncertainty or exploitable row structure. It is not a claim that every uncertain rigid-body model has a tractable exact vertex description.

Execution tightening: Suppose a row acts on a model quantity $y(a; \theta)$ and the realized quantity is $y_{\text{real}} = y(a; \theta_{\text{true}}) + \delta$. This quantity can be a wrench or an acceleration from which a wrench is computed. Tighten

$$L_i(\theta)y \leq b_i(\theta) \quad \text{to} \quad L_i(\theta)y \leq b_i(\theta) - \varepsilon \|L_i(\theta)\|_*. \quad (23)$$

Here $\|\cdot\|_*$ is the dual of the discrepancy norm.

Theorem 2 (Conditional executed-wrench certificate). *Assume $\theta_{\text{true}} \in \Theta$, and suppose the final command belongs to the nonempty tightened robust set. Algorithm 1 ensures this through a joint projection, or through channel-wise projection and convex reaction averaging when that robust set is a product. Then:*

- 1) *The model quantity evaluated with θ_{true} satisfies every tightened row.*
- 2) *If $\|\delta(\tau)\| \leq \varepsilon$ throughout the control period, the realized quantity satisfies every original row throughout that period.*
- 3) *If only $\|\bar{\delta}\| \leq \varepsilon$ is known for the period average, only the period-averaged realized quantity is certified by this argument.*

Proof. Membership of the final command in the tightened robust set gives the model statement because $\theta_{\text{true}} \in \Theta$. For each row, Hölder's inequality gives $L_i\delta \leq \|L_i\|_* \|\delta\| \leq \varepsilon \|L_i\|_*$. Adding this inequality to the tightened model row proves the pointwise statement. The same calculation with $\bar{\delta}$ proves the average statement. $\square$

For (19), both channel sets must include their required robust rows and tightening. The first-step reactions are then averaged within the same tightened convex reaction set. If parameter uncertainty introduces coupled rows or parameter-dependent channel maps that do not admit one common product representation, the joint final projection is required. Separation of the nominal model alone does not establish robust separability.

Vertex reduction remains exact after tightening if the tightened residual is separately convex in the parameters. In particular, if the original residual has this property, $L_i(\theta)$ is separately affine, and one fixed ε applies over the parameter box, then adding $\varepsilon \|L_i(\theta)\|_*$ preserves separate convexity. A parameter-dependent error bound requires its own convexity check.

A period-averaged wrench certificate does not prohibit a short force spike, contact loss, slipping, or tipping by itself. Those outcomes depend on the intervening dynamics and validity of the contact model. The transport experiments consequently report both averaged and instantaneous diagnostics and distinguish observed tipping counts from the theorem's conclusion.

Set-membership adaptation: Let a valid measurement window satisfy

$$y_k = \Psi_k \phi + e_k, \quad |e_k| \leq \bar{e}_k, \quad \theta = \eta(\phi). \quad (24)$$

The bound is componentwise and must include sensor error, model discrepancy, and regressor error. If a regressor is uncertain, its contribution must be bounded uniformly over the current parameter set, rather than evaluated at a point estimate.

Proposition 5 (Parameter containment). *Suppose $\phi_{\text{true}} \in \hat{P}_0$ and $\theta_{\text{true}} \in \Theta_0$. In a valid window, let $\hat{P}_k$ contain*

$$\hat{P}_{k-1} \cap \{\phi : |y_k - \Psi_k \phi| \leq \bar{e}_k\}, \quad (25)$$

and otherwise retain $\hat{P}_k = \hat{P}_{k-1}$. Set

$$\Theta_k = \Theta_{k-1} \cap \square(\hat{P}_k), \quad (26)$$

where $\square$ is an outer box enclosure on a domain where η is defined. If each accepted window obeys (24), then $\phi_{\text{true}} \in \hat{P}_k$ and $\theta_{\text{true}} \in \Theta_k \subseteq \Theta_{k-1}$ for all k .

Proof. Inductively, the true parameter lies in the previous set and satisfies the new measurement inequalities, hence it belongs to their intersection and its outer enclosure. Its image therefore belongs to both sets in (26). $\square$

At a fixed robot state, decreasing Θ enlarges the robust feasible set when all other constraints and error bounds are fixed. This is a reduction of parameter conservatism, not a recursive-feasibility proof along a changing trajectory. An empty measurement intersection indicates inconsistent assumptions; the implementation rejects and counts that update. Such rejection preserves the previous set only if its containment assumption was already valid.

Application to tray support: Let n be the tray normal and let $u_1, \dots, u_8$ be eight in-plane directions. A box has base half-width r , center-of-mass height h , and friction coefficient μ . With $c = \cos(\pi/8)$ and $s = a_B - g$, where a_B is the tray-point acceleration, use

$$\begin{aligned} h u_j^\top s &\leq r c n^\top s & (\text{support}), \\ u_j^\top s &\leq \mu c n^\top s & (\text{friction}), \\ n^\top s &\geq 3 \text{ m/s}^2 & (\text{normal load}). \end{aligned} \quad (27)$$

The octagonal support region lies inside the square base. These rows are affine in joint acceleration and separately affine in h or μ , so their parameter endpoints suffice. A level-tray virtual equality supplies the assumed orientation. These are support *motion* rows, not the reaction channel of (2).

To identify h without measuring the object's absolute position, integrate wrist force and torque over a window. In a sticking model, the force impulse Y_F and torque impulse Y_τ satisfy

$$Y_F = m\nu, \quad Y_\tau = (mh)n \times \nu + m d \times \nu, \quad (28)$$

where ν is the gravity-compensated change in tray-point velocity and d is the object offset. Direct estimation in (m, mh) must bound the d term. If the offset is tracked, its uncertainty can be reduced.

Alternatively, the center of pressure (COP) is $p_c = d - h\kappa$, with $\kappa = \nu_t/\nu_n$. Differencing two windows yields

$$\Delta p_c = -h \Delta \kappa + \Delta d. \quad (29)$$

The constant offset cancels. COP measurement errors, uncertain regressors, and a bounded creep Δd form the measurement-error bound. The resulting scalar interval update is performed while executing the transport trajectory. It does not guarantee excitation, and it may provide little contraction when the task is uninformative. Friction remains at its prior in the present study because the friction rows do not bind.

IV. EXPERIMENTAL EVALUATION

The evaluation comprises three MuJoCo simulation studies and a separate computational benchmark. Writing tests physical–virtual equality assignment and the motion–reaction update; dual-arm transport tests calibration-induced loading and constraint tradeoffs; nonprehensile transport tests robust support constraints and adaptation. Each task presents its relevant comparisons with its results. All physical quantities reported here are simulator measurements.

The arms use Franka-like kinematics and inertial parameters derived from Gaz *et al.* [25], with 0.1 kg m^2 joint armature. The batched model was checked against the configured MuJoCo model [26] for mass matrices, bias forces, and Jacobians. PR-MPPI and CAMP-MPPI are reimplemented from the cited formulations. Other comparison mechanisms and their adaptations are documented in Appendix A-F.

The evaluated UP-MPPI controller uses Algorithm 1. Writing and the constrained dual-arm variant use raw motion averaging and projected-reaction averaging; the waiter has only motion rows and uses a projected raw mean. PR and CAMP retain projected-sample averaging. MPPI variants within a task share the stated sample count and horizon; order-1 and order-2 exploration have different units. Simulation advances at the prescribed physics and control steps, independently of wall-clock planning duration. Section IV-D measures computation separately.

Performance tables give seed means and, where reported, descriptive sample standard deviations. No-tip counts, constraint residuals, and execution-bound checks are distinct measurements; the small seed counts are not failure-probability estimates.

A. Surface writing

This task tests physical–virtual equality assignment and the motion–reaction update. A 7-DoF arm traces a $16 \times 8 \text{ cm}$ lemniscate on a spherical board of radius 0.6 m in 8 s , with a requested reaction band of $[2, 4] \text{ N}$. The board supplies the normal reaction; tool orientation is a virtual objective. CAMP-MPPI uses an orientation cost, whereas UP-MPPI imposes hard virtual rows. A separate ablation treats orientation as physical; it is not standard CAMP-MPPI.

MPPI runs at 100 Hz with $H = 15$ and $K = 256$. Physics and low-level inverse dynamics run at 1 kHz . The PR-MPPI tracker uses gains $K_P = \omega_{\parallel}^2$, $K_D = 2\omega_{\parallel}$ with $\omega_{\parallel} = 2\pi \times 10 \text{ rad/s}$; 10 Hz is the nominal natural frequency, not its execution rate. Velocity and acceleration exploration scales are 0.1 rad/s and 0.8 rad/s^2 , respectively. Contact stiffness, damping, and friction are $2 \times 10^4 \text{ N/m}$, 60 N s/m , and 0.2 . Joint viscous friction of 1 N m s/rad is unmodeled. Order-2 controllers use HOCBF joint limits, velocity rows, and a velocity cost for task-null-space damping.

The board-model offsets are $\delta \in \{-1, -0.5, 0, 0.5, 1\} \text{ mm}$, with three seeds per condition. Force and position metrics exclude the first 0.5 s ; orientation RMS covers the executed trajectory. No execution tightening is applied in writing; the reaction constraint is imposed on the command, and measured force occupancy is an empirical outcome.

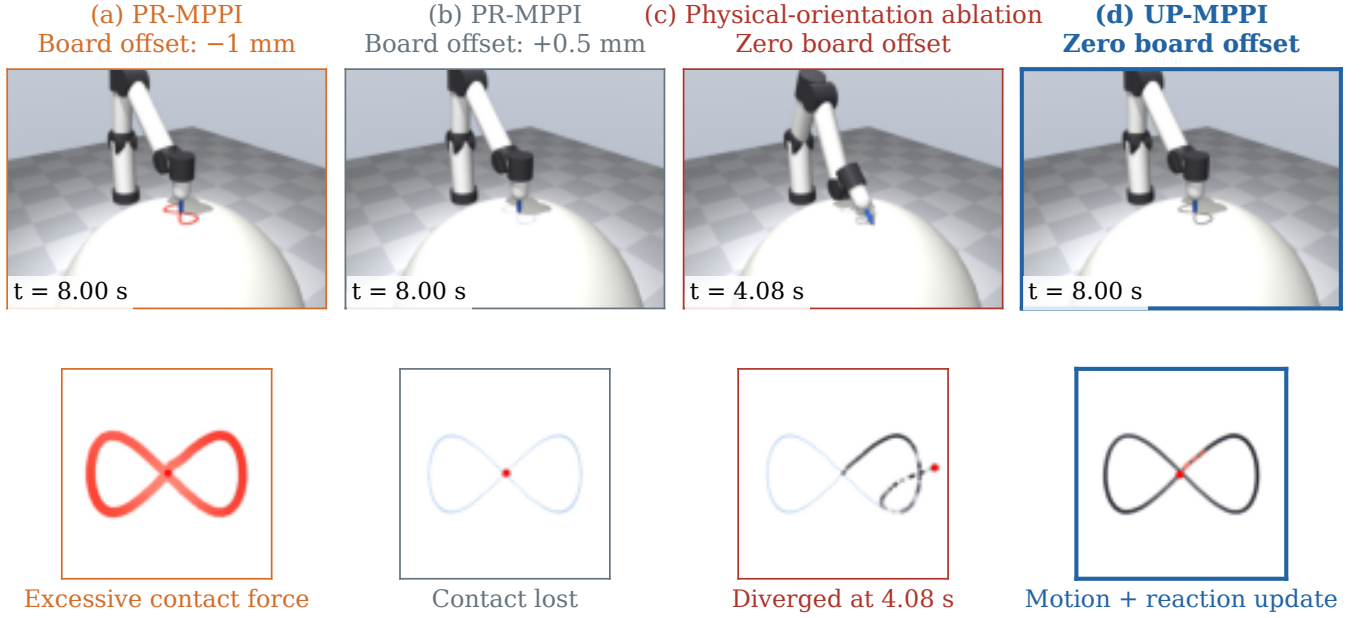


Fig. 3. Writing snapshots and contact close-ups, seed 0. Columns (a), (b), and (d) show 8 s; the physical-orientation ablation (c) terminates at 4.08 s. Ink is black for measured force in $[2, 4]$ N, red above 4 N, and gray below 2 N; the thin blue curve is the reference. The lower row hides the arm. Conditions are labeled in each panel; the orientation ablation is distinct from standard CAMP-MPPI.

TABLE II
WRITING AT ZERO BOARD OFFSET, THREE SEEDS. TRACKING AND ORIENTATION ERRORS ARE RMS; FORCE IS A TEMPORAL MEAN. METRICS FOR DIVERGENT RUNS COVER ONLY THEIR EXECUTED PORTIONS.

Controller	Force [N]	In band [%] $\uparrow$	Tracking [mm] $\downarrow$	Tilt [deg] $\downarrow$	Completed
PR-MPPI	0.00	0.0	1.09	0.00	3/3
Anchored PR-MPPI	0.00	0.0	1.09	0.00	3/3
CAMP-MPPI (orientation cost)	3.04	99.1	1.77	0.72	3/3
Orientation-as-physical ablation	1.91	42.1	20.99	25.54	0/3
UP-MPPI (mixed update)	3.03	99.3	1.58	0.11	3/3
UP-MPPI, joint projection	3.03	99.3	1.58	0.11	3/3
UP-MPPI, $W = I$	3.28	16.8	1.87	0.18	3/3

The UP-MPPI rows use a raw motion mean and averaged projected reactions, without execution tightening. Measured force-band occupancy is an empirical outcome. Blue rows identify the proposed inertia-metric implementations. Bold marks the best displayed in-band, tracking, and tilt values among completed runs (ties included); PR loses contact at this offset, so its small pose errors do not indicate successful writing. **Compared with CAMP-MPPI: 11.0% lower tracking RMSE and 85.1% lower tilt**, with 99.3% versus 99.1% force-band occupancy.

Equality assignment and channel update: Figure 3 and Table II show the consequences of the constraint assignment. Treating orientation as physical causes all three zero-offset runs to diverge, after 4.1 s on average: the board cannot supply the modeled orienting reaction. CAMP-MPPI completes the task with an orientation cost; hard virtual rows reduce tilt from 0.72° to 0.11° without an orientation penalty weight.

The mixed update commands $[2.12, 3.39]$ N and produces a measured mean of 3.03 N. Force-band occupancy is 99.3%, compared with 99.1% for CAMP-MPPI, while tracking RMSE falls from 1.77 to 1.58 mm. Thus the improvement over CAMP is principally in path and orientation, with comparable force regulation. The measured force traces are shown in Fig. 4(a).

The separable and joint inertia-metric solvers match to 2.4×10^{-11} N on stored force traces across all offsets and seeds. Replacing M by I raises tracking error to 1.87 mm and reduces band occupancy to 16.8%. This ablation retains the reaction replacement after a Euclidean joint projection, whose correction can already have changed the motion. It supports

the role of separation in this task. The computational effect of using the equivalent separable solver is measured separately in Section IV-D.

Model offsets and additional controls: At $\delta = -1$ mm, PR-MPPI produces 15.97 N against the 15.94 N prediction from (13) and (14); at -0.5 mm the values are 7.98 and 7.97 N. Contact is effectively lost at nonnegative offsets. Anchoring removes offset-induced loading but also removes sustained writing contact. Constant board offsets do not change the contact Jacobian, so the order-2 controllers have identical results across this offset sweep (Fig. 4(b)).

A one-step task-space QP with the same relevant hard rows achieves 1.00 mm RMSE and 100.0% occupancy, outperforming UP-MPPI on this simple trajectory. This result and the full penalty/filter sweep are retained in Appendix A-F, Table VIII. That appendix also reports contact-task adaptations of safe-sampling methods and their formulation differences. The present task establishes the effects of the contact model, metric, and channel update; the common transport task below supplies

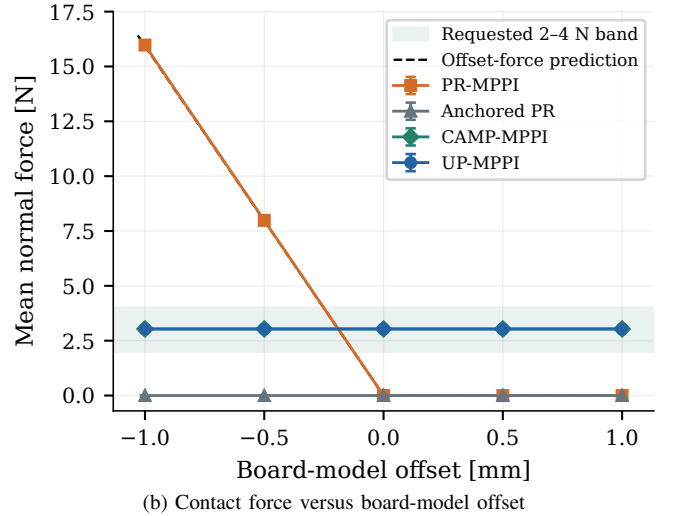
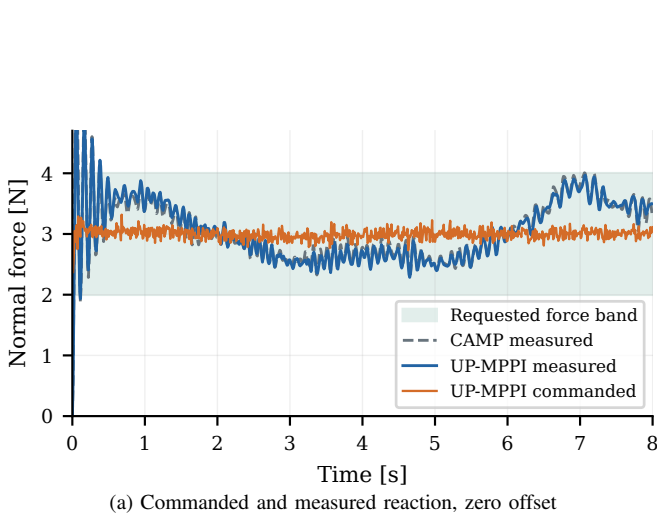


Fig. 4. Writing force measurements. (a) Seed 0; the shaded region is the requested band. Traces are decimated to 10 ms, with startup transients shown; table metrics exclude the first 0.5 s. (b) Three-seed means with sample-standard-deviation whiskers. The dashed curve is the series-stiffness prediction using the executed tracker’s effective stiffness. Both panels concern measured force without execution tightening.

TABLE III

DUAL-ARM JOINT-LIMIT STRESS TEST AT ZERO CALIBRATION OFFSET, MEANS OVER THREE SEEDS. FORCE IS THE INTERNAL-FORCE NORM; CHAIN ERROR IS TRANSLATIONAL.

Controller	Violation	Chain	Tilt	Force
PR-MPPI	0.000	0.05	0.03	0.47
PR without retraction	0.000	0.20	3.61	0.53
PR without inequalities	0.059	0.05	0.03	0.50
UP-MPPI, tightened	0.000	0.67	1.76	2.00
Reduced-constraint variant	0.008	0.54	1.70	1.08
Shield adaptation	0.000	0.35	1.70	1.41
DualGuard adaptation	0.000	0.50	1.69	1.52
GS raw-mean adaptation	0.002	0.35	1.70	1.09

Units: bound violation [rad], chain error [mm], tray tilt [deg], internal force [N]. The last four rows use no execution tightening or reaction box; they do not isolate the margin alone. Blue identifies the proposed controller; bold marks the best displayed value in each metric, including ties. Constraint configurations differ, so this is not an overall method ranking.

the main comparison with external sampling mechanisms.

B. Dual-arm transport

Two Franka-like arms, spaced 0.7 m apart, carry a 1 kg tray through compliant six-dimensional grasps. Six relative-pose equalities are physical; tray roll and pitch are virtual. Translational and rotational grasp stiffnesses are 3×10^4 N/m and 300 N/m/rad. Physics runs at 2 kHz and MPPI at 50 Hz, with $H = 15$ and $K = 256$.

The tray is lowered from (0.60, 0, 0.40) to (0.42, 0, 0.20) m. An artificial joint-4 lower limit is 0.06 rad above its inverse-kinematics goal, with a 0.05 rad safety margin. The modeled right-arm base is shifted by $\delta_b \in \{0, 0.5, 1, 2\}$ mm along y . UP-MPPI uses a reaction box of 5 N and 1 N/m and motion rows tightened by per-joint acceleration-discrepancy bounds calibrated on separate runs. Internal-force means exclude the first 1 s; bound violations are checked over the full run.

Calibration-induced loading: Figures 5 and 6 show how a geometric offset becomes internal force. PR-MPPI rises from

0.47 to 15.23 N as the base error reaches 2 mm. Anchored PR gives 0.45 N at that offset, while tightened UP-MPPI gives 2.00 N. These results support the physical-manifold sensitivity argument. Anchored PR attains the lowest force, and PR also retains better tray levelness at zero offset (0.03° versus 1.76°).

Constraint and goal-approach tradeoffs: Table III reports the zero-offset joint-limit test. Removing PR retraction raises the modeled command-equality residual to 0.111; removing inequality projection permits 0.059 rad of bound violation. The reduced-constraint order-2 variant violates the bound by 0.008 rad. It removes both tightening and the reaction box, also disabling clamped-reaction averaging, so it is not a single-factor margin ablation.

Tightened UP-MPPI has no measured bound violation in the three zero-offset runs and stops 0.261 rad from the joint-space goal, versus 0.115 rad for PR. Across the calibration sweep its largest per-joint discrepancy is 0.67 times the corresponding calibrated bound. Shield and DualGuard adaptations maintain the joint bound while reaching approximately 0.11 rad from the goal; the GS raw-mean adaptation violates it by 0.002 rad. Those adaptations omit the tightening and reaction box used by UP-MPPI (Appendix A-F). The comparison therefore documents the cost of the implemented constraint configurations alongside their goal approach.

C. Uncertain nonprehensile transport

This task evaluates robust support constraints, the motion update, and online parameter adaptation. A box with base half-width $r = 0.02$ m and height 0.24 m stands on a tray. Its true mass, CoM height, and friction coefficient are 0.6 kg, $h = 0.14$ m, and $\mu = 0.6$. The prior is $h \in [0.05, 0.18]$ m and $\mu \in [0.4, 0.8]$. The tray moves 0.4 m in 0.8 s, holds for 1.2 s, and returns. Requested peak acceleration is 3.6 m/s^2 , above the true support limit of approximately 1.4 m/s^2 along a base axis.

Physics runs at 1 kHz with compliant contact and an elliptic friction cone; control runs at 50 Hz with $H = 20$ and $K = 256$. Three virtual rows hold tray levelness and heading. Execution

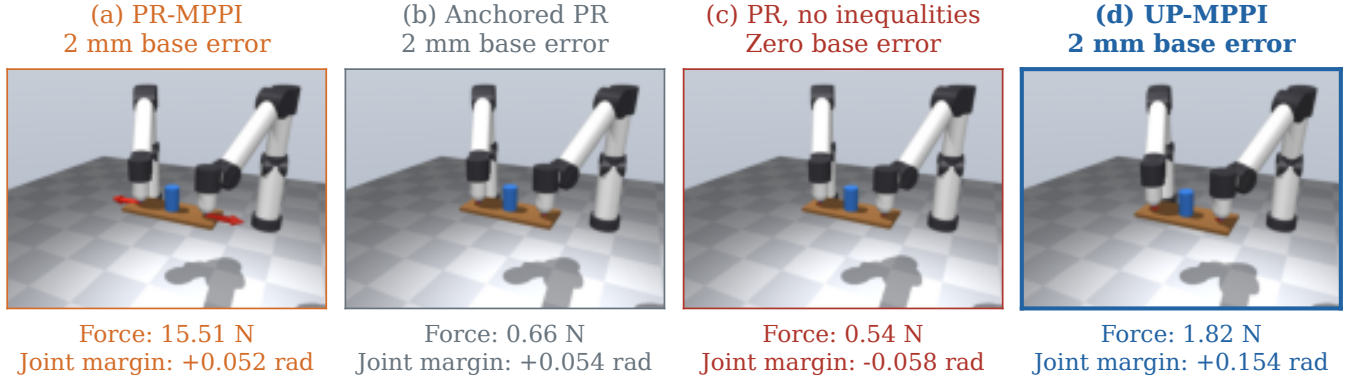


Fig. 5. Dual-arm final states at 6 s, seed 0. Force arrows have a common scale. Labels give instantaneous internal-force norms and the smaller joint-4 margin across both arms. Columns (a), (b), and (d) use 2 mm base error; the inequality ablation (c) uses zero error. UP-MPPI additionally imposes the specified virtual rows, reaction box, and calibrated motion tightening.

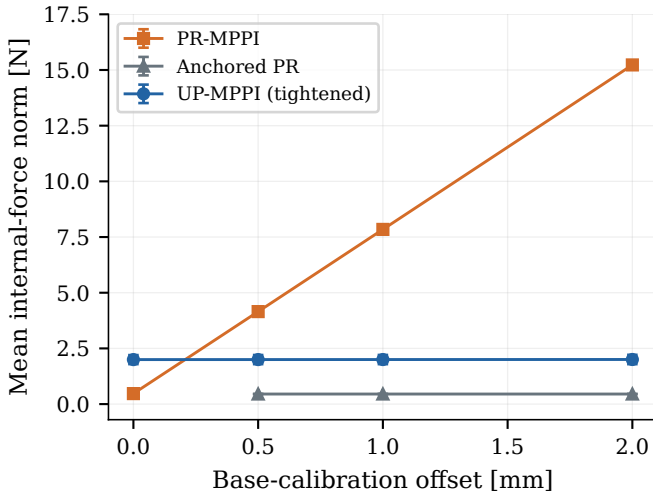


Fig. 6. Mean internal-force norm versus base-calibration error, three seeds per point. Whiskers show sample standard deviations where visible. Anchoring reduces the calibration load; UP-MPPI enforces additional motion and reaction constraints.

compensates modeled damping and the load measured at the preceding physics step. All variants share the task cost. Nominal control assumes $h = 0.07$ m, robust control keeps the prior fixed, and the oracle knows the true parameters. The adaptive variants use direct F/T impulses, COP differences, or tracked object position.

F/T estimates use 100 ms measurement windows after settling. Separate calibration runs determine measurement bounds and the period-averaged execution bound $\varepsilon = 0.165$ m/s². The latter is 1.5 times the largest calibration deviation, 0.110 m/s². Sensor noise, object-offset bounds, creep bounds, and the update gate are specified in Appendix A-F. The octagon inside the square footprint supplies a geometric reserve distinct from this calibrated execution margin.

Robustness and adaptation: The nominal model tips the box in all five main runs. Robust, adaptive, and oracle control have no tips. Robust move-2 RMSE is 41.2 mm, versus 27.1 mm for the oracle. COP-difference adaptation reaches 30.4 mm: 26.1% below the fixed robust prior and 76.6% of

the robust-to-oracle gap recovered (Table IV). Its final interval union is $[0.131, 0.149]$ m and contains the true height. The direct F/T interval remains wider, $[0.106, 0.174]$ m, because offset uncertainty dominates its torque bound. Figures 7 and 8 show the corresponding motion and interval evolution.

A sweep at $h \in \{0.08, 0.11, 0.17\}$ m uses three seeds per condition. Including the main study, fixed-robust, adaptive, and oracle variants have zero tips in 61 runs. COP adaptation closes approximately 98%, 100%, 77%, and 0% of the move-2 robust-to-oracle gap at the four heights. Near the upper prior endpoint, the robust rows already leave little room for improvement. The RLS point-estimate comparator reaches 27.2 mm on move 2 and has no tips in its 28 main-and-sweep runs across two initial estimates. The distinction of set membership here is conditional parameter containment; RLS has better measured tracking in this study.

Motion-update ablation: Table V compares raw and projected averaging, penalty rollouts (A1), penalty rollouts with a final filter (A2), a one-step hard-constrained QP (A3), and the RLS estimator (B1). Their definitions and tuning rules are in Appendix A-F. With the same explicit margin, raw averaging reduces move-1 RMSE from 62.4 to 39.2 mm and move-2 RMSE from 57.2 to 41.2 mm. Without tightening, raw robust control reaches 30.2 mm on move 1, but its realized COP reaches the edge in the height sweep. This isolates averaging-induced inward displacement from the chosen execution margin.

The best tightened A2 gives 48.2 and 46.5 mm, compared with 39.2 and 41.2 mm for projected rollouts: reductions of 18.8% and 11.4%. A1 tracks well at a large penalty but exceeds the true support rows at $h = 0.17$ m (ratio 1.09). The best tested A3 gain gives 76.5 and 80.3 mm on the main task. Unlike the writing penalty/filter comparison, this motion-only experiment isolates support-projection placement without changing a reaction-averaging channel.

External comparison mechanisms: Table VI evaluates local adaptations of Shield-MPPI [13], DualGuard MPPI [15], and GS-MPPI [14], together with offline robust planning after Heins and Schoellig [22] and online robust NMPC. Sampling methods share the robot model, task cost, horizon, and exploration. These are task adaptations: in particular, the

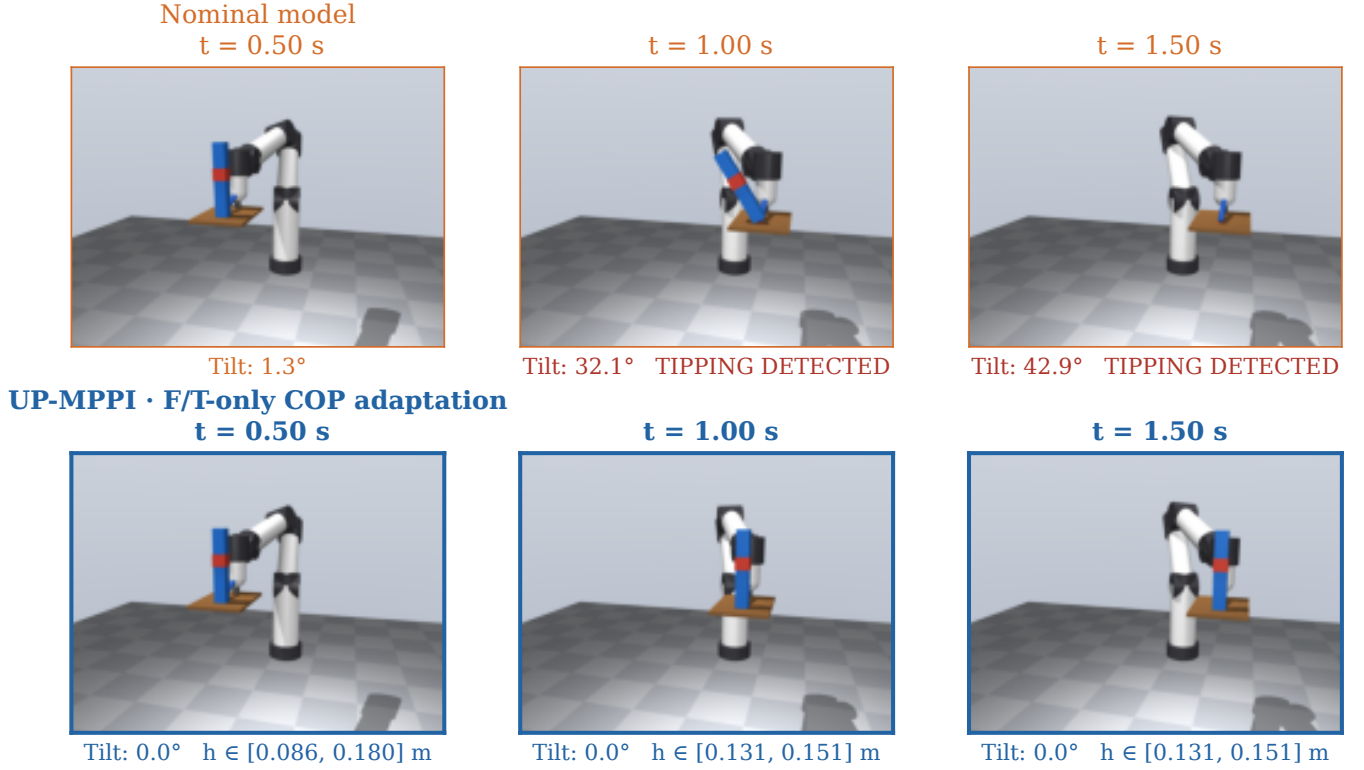


Fig. 7. Uncertain-payload transport, seed 0, true CoM height $h = 0.14$ m. The nominal controller assumes $h = 0.07$ m and the load tips during the first move. F/T-only COP adaptation maintains an upright load in this run and contracts the height interval. Each panel shows a recorded simulator state at the labeled time. Table IV aggregates all five main seeds.

TABLE IV

TRANSPORT AT $h = 0.14$ M. RMSE IS MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER FIVE SEEDS; INTERVALS ARE UNIONS OF FINAL PARAMETER INTERVALS. TIPPED NOMINAL RUNS HAVE NO FULL-MOTION RMSE.

Controller	Tips	Move 1 [mm]	Move 2 [mm]	Final h [m]
Nominal	5/5	—	—	—
Robust	0/5	39.2 ± 0.6	41.2 ± 1.3	$[0.050, 0.180]$
Direct F/T	0/5	38.0 ± 0.7	39.3 ± 1.1	$[0.106, 0.174]$
COP differences	0/5	39.1 ± 0.8	30.4 ± 0.6	$[0.131, 0.149]$
Object tracked	0/5	28.0 ± 0.7	29.8 ± 0.6	$[0.131, 0.149]$
Oracle	0/5	25.7 ± 0.2	27.1 ± 0.5	$[0.140, 0.140]$

Blue identifies fixed-robust UP-MPPI and its F/T-only COP adaptation; bold marks the lowest displayed tracking error, including the oracle. **COP adaptation reduces move-2 RMSE by 26.1% relative to the fixed robust prior**, recovering 76.6% of the robust-to-oracle gap.

GS variant applies a soft minimum to input residuals and executes a filtered raw mean at the planning rate, whereas the source selects a best sampled input and uses a faster inner filter. Appendix A-F gives all formulation differences and selection rules. The offline planner has full-reference preview; the optimizers use different discretizations and slack treatment.

With the same numerical tightening, the GS adaptation nearly matches UP-MPPI on move 1 (39.3 versus 39.2 mm); Shield gives 53.6 mm and DualGuard 110.2 mm. DualGuard's maximum period-averaged discrepancy is 0.257 m/s^2 , above $\varepsilon = 0.165 \text{ m/s}^2$, despite no tips. The same numerical tightening therefore needs a separate discrepancy check for each controller.

The offline plan has the smallest main-task tracking errors: move 1 is 23.1 and 27.7 mm without and with tightening. Its untightened feedback execution exceeds the model support boundary at $h = 0.14$ m (ratio 1.02) and tips at $h = 0.17$ m

(ratio 1.24). Online NMPC gives 34.5 and 45.1 mm, versus 30.2 and 39.2 mm for UP-MPPI, with no solver failures. Computation is assessed separately in Section IV-D.

Execution-certificate diagnostics: Across the 61 robust/adaptive/oracle runs, the largest true-parameter model support ratio is 0.926, the largest period-averaged acceleration discrepancy is 0.115 m/s^2 , and the largest averaged COP-to-edge ratio is 0.869. All adaptive intervals contain the true height, with no rejected inconsistent measurements. These checks support the averaged statement of Theorem 2 for the evaluated runs.

The pointwise bound is not verified: a diagnostic oracle run exceeds ε in 1.6% of settled physics substeps, and instantaneous COP touches an edge in at least one substep of 60 of the 61 runs. Accordingly, the zero-tip counts remain empirical outcomes rather than consequences proved by the period-

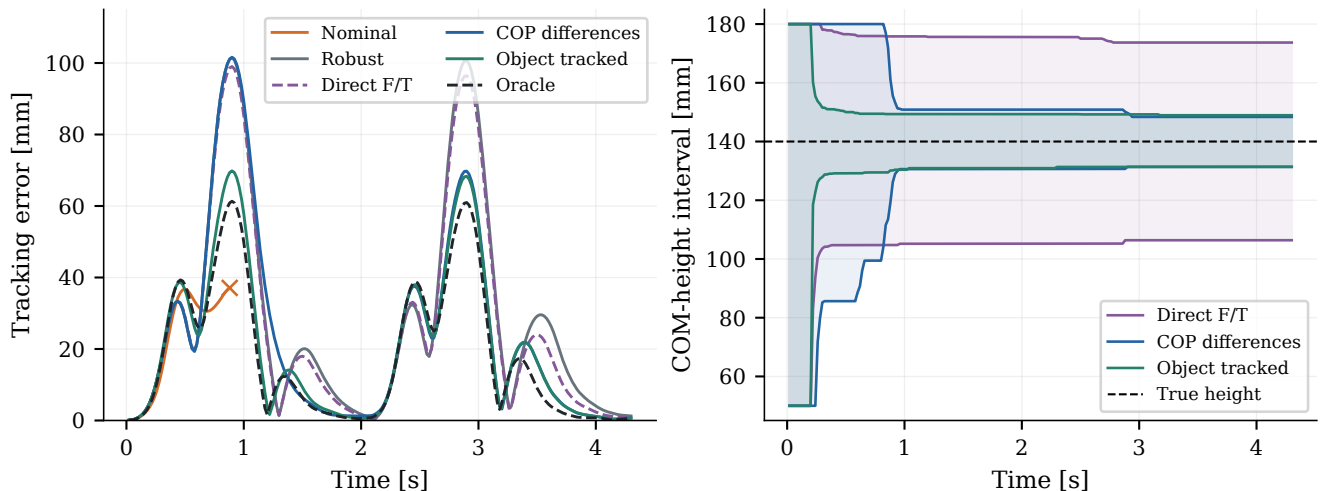


Fig. 8. Transport at $h = 0.14$ m, seed 0. Left: tray-point tracking error, with nominal execution truncated at tipping. Right: CoM-height intervals; the dashed black line is the true height. COP differences reduce the effect of an unknown object offset using F/T measurements alone.

TABLE V
MATCHED TRANSPORT BASELINES AND AVERAGING ABLATIONS AT $h = 0.14$ M (FIVE SEEDS). MODEL RATIO USES TRUE PARAMETERS; COP RATIO IS PERIOD-AVERAGED. A RATIO ABOVE ONE VIOLATES THE CORRESPONDING BOUNDARY.

Controller	Tips	Move 1 [mm]	Move 2 [mm]	Model ratio	COP ratio
Robust raw, tightened	0/5	39.2 ± 0.6	41.2 ± 1.3	0.66	0.63
Robust projected, tightened	0/5	62.4 ± 0.6	57.2 ± 1.9	0.60	0.57
Robust projected, no tightening	0/5	46.8 ± 0.4	47.3 ± 1.6	0.73	0.69
Robust raw, no tightening	0/5	30.2 ± 0.2	31.9 ± 0.5	0.78	0.77
A1 penalty, $w_s = 10^5$	0/5	36.2 ± 0.2	37.3 ± 0.6	0.90	0.83
A2 penalty + filter, tightened	0/5	48.2 ± 0.7	46.5 ± 1.3	0.66	0.62
A3 one-step QP, $\omega = 6$	0/5	76.5 ± 0.0	80.3 ± 0.0	0.78	0.75
B1 RLS, $h_0 = 0.115$ m	0/5	27.3 ± 0.4	27.2 ± 0.5	0.88	0.85
B1 RLS, $h_0 = 0.070$ m	0/5	28.4 ± 0.7	27.2 ± 0.5	0.90	0.86

Tightened rows use $\varepsilon = 0.165 \text{ m/s}^2$. The same numerical margin is not an unconditional physical-safety guarantee; it requires the execution-discrepancy bound. Blue identifies the proposed motion-only update with tightening. **Compared with projected averaging at the same margin: 37.3% lower move-1 RMSE and 28.1% lower move-2 RMSE.** Rows with different margins or parameter information are not ranked together.

averaged certificate.

D. Computational performance

This benchmark measures the cost of equivalent separable and joint projections. The trajectory studies above use $K = 256$; here we sweep $K \in \{256, 1024, 4096\}$ at fixed task benchmark states, with $H = 15$ for writing and $H = 20$ for transport. The reference states and measurement procedure are independent of the trajectory-level performance averages. Thus a time reported at $K = 1024$ is a separate scaling measurement, not the recorded latency of the $K = 256$ performance runs.

Implementation and measurement scope: An NVIDIA Warp float32 kernel assigns one sample to each thread and executes the full rollout. Device operations include model evaluation, row construction, dual active-set projection, integration, scoring, weights, and the raw mean. For writing, final filtering reuses the kernel with one thread and zero exploration; CPU reaction reconstruction and its device-to-host transfers complete the command. *Planning time* includes this path. *Rollout-kernel time* is a separate synchronized measurement of the sampled rollouts only.

The device is an RTX 4060 with 8 GB memory. After ten warm-up plans, each GPU configuration is measured in three rounds of fifty plans; Table VII reports the smallest round median. Kernel time uses a separate fifty-plan pass. The NumPy reference is the median of three plans. Other CPU jobs ran on the workstation, so these measurements describe the tested implementation and benchmark conditions.

Separation and sample-count scaling: At $K = 1024$, writing takes 4.43 ms with separation and 7.64 ms with a joint projection: 42.1% less planning time. Kernel time falls from 2.817 to 1.241 ms, a 55.9% reduction. Both variants use the same mixed update and reconstruction path, and their stored contact-force traces match as reported in Section IV-A. Robust transport takes 8.65 ms; it has no separate reaction channel and is not a test of separation speedup.

The $K = 1024$ measurements are below the nominal writing and transport periods of 10 and 20 ms. At $K = 4096$, joint writing takes 10.89 ms and exceeds its period. These are planner measurements, not complete hardware control-loop latency or deadline-miss statistics.

For context, online NMPC records a mean CPU step time of 436–465 ms during its transport trajectories. That trajectory

TABLE VI

TASK-ADAPTED SAFE-SAMPLING AND OPTIMIZATION BASELINES AT $h = 0.14$ M. FIVE SEEDS FOR SAMPLING METHODS; ONE DETERMINISTIC RUN FOR EACH OPTIMIZER. MODEL/COP RATIOS ARE MAXIMA ACROSS RUNS; RMSE VALUES ARE MEANS.

Controller	Tips	Move 1 [mm]	Move 2 [mm]	Model	COP	Error [m/s ²]
<i>No execution tightening ($\varepsilon = 0$)</i>						
UP-MPPI	0/5	30.2	31.9	0.78	0.77	0.079
Shield adaptation ($C = 1000$)	0/5	40.5	40.2	0.78	0.72	0.065
DualGuard adaptation	0/5	85.4	114.1	0.60	0.59	0.204
GS raw-mean adaptation ($\rho = 100$)	0/5	30.2	32.1	0.78	0.79	0.078
Offline robust plan	0/1	23.1	22.2	1.02	0.96	0.108
Online robust NMPC ($w_j = 0.1$)	0/1	34.5	37.1	0.78	0.74	0.060
<i>Same numerical tightening ($\varepsilon = 0.165$ m/s²)</i>						
UP-MPPI	0/5	39.2	41.2	0.66	0.63	0.063
Shield adaptation ($C = 1000$)	0/5	53.6	49.5	0.66	0.61	0.055
DualGuard adaptation	0/5	110.2	237.7	0.52	0.48	0.257
GS raw-mean adaptation ($\rho = 20$)	0/5	39.3	41.4	0.66	0.63	0.059
Offline robust plan	0/1	27.7	26.7	0.85	0.83	0.109
Online robust NMPC ($w_j = 0.1$)	0/1	45.1	46.1	0.65	0.64	0.052

These are local adaptations, including a raw-mean GS variant, not original author implementations. Equal numerical tightening does not establish equal certificates: the DualGuard adaptation exceeds the execution-error bound. Blue identifies UP-MPPI; bold marks the lowest displayed tracking error separately for each margin. Red flags a model ratio above one or a discrepancy above the tightened bound; model/COP ratios are diagnostics, not tracking objectives.

TABLE VII

PLANNING TIME [MS], WITH ROLLOUT-KERNEL TIME IN PARENTHESES. GPU VALUES ARE THE SMALLEST ROUND MEDIAN AFTER WARM-UP, ON AN RTX 4060. NUMPY USES THE MEDIAN OF THREE PLANS. WRITING INCLUDES CPU REACTION RECONSTRUCTION AND ASSOCIATED TRANSFERS.

Controller	NumPy $K = 256$	GPU $K = 256$	GPU $K = 1024$	GPU $K = 4096$
Waiter, robust	293.2	8.68 (6.97)	8.65 (7.03)	17.30 (15.57)
Writing, separable	183.9	3.85 (0.72)	4.43 (1.24)	6.92 (1.95)
Writing, joint	199.8	6.46 (2.43)	7.64 (2.82)	10.89 (4.03)

Blue identifies the separable writing implementation. Bold compares only separable and joint writing at the same sample count. **At $K = 1024$: 42.1% less planning time and 55.9% less rollout-kernel time**, with matching stored contact-force traces.

average differs from the GPU fixed-state benchmark in device, computational path, and aggregation. It is reported separately and is not used to form an algorithmic speedup ratio. The direct efficiency result is the same-device comparison of equivalent writing projections.

V. DISCUSSION AND CONCLUSION

UP-MPPI connects geometric constraints, contact reactions, and uncertain support wrenches through a common projection framework. The separation condition identifies when motion and reaction can be corrected independently. This structure supports a raw motion mean and an averaged projected reaction in the same controller, while explicit tightening supplies an execution margin distinct from the inward displacement caused by sample clipping.

The experiments test these contributions at different levels. Writing demonstrates physical–virtual equality assignment and improved path and orientation tracking while retaining CAMP-like force regulation. Dual-arm transport verifies calibration-induced loading and exposes the goal-approach cost of the chosen constraint configuration. The waiter study isolates the motion update and shows how F/T-only adaptation recovers performance from a fixed robust prior. External comparison mechanisms are evaluated on this shared transport problem; the GS adaptation attains similar tracking, while the proposed framework also supplies the contact-specific structure studied in writing. Equivalent separable and joint writing projections

establish the computational benefit independently of those cross-method comparisons.

The evidence supports a structured controller construction rather than a universal performance ordering. The one-step QP is strongest on the simple writing path, RLS tracks the excited transport motion better than the interval estimator, and full-reference offline planning has the lowest main-task tracking errors. Their outcomes remain in the task comparisons. Set membership contributes conditional parameter containment, and the execution analysis makes the required discrepancy bounds explicit.

The principal scope conditions concern contact validity and error bounds. The rigid-contact model is local to its contact mode; calibrated measurement and execution bounds are assumptions, and COP gating is not a complete slip detector. Parameter containment does not ensure excitation or recursive feasibility along a changing trajectory. The transport data support period-averaged wrench constraints, while instantaneous deviations exceed the selected bound; zero tipping is therefore an observed outcome. The fixed-temperature sampling characterization is separate from the adaptive-temperature implementation. All present evidence is simulation and computation; real-robot execution, sensor calibration, and control-stack timing remain to be evaluated.

The resulting framework explains which projections can be decomposed, when coupled constraints require a joint solve, and how uncertainty enters the commanded and realized wrench claims. These distinctions organize the geometry,

channel update, and adaptation within one sampled predictive controller.

APPENDIX A SUPPORTING ANALYSIS

A. Finite-step barrier layer

Continuous-time barrier rows do not by themselves establish safety of a finite-step implementation [27]. The following bound concerns the semi-implicit rollout model used here.

Lemma 1. *Suppose (4) is enforced without relaxation at every rollout step, $0 < k_i \Delta t \leq 1$, and $\|\nabla^j h\| \leq L_j$ for $j = 1, 2, 3$ on the step segments. Let $\|\dot{q}_t\|, \|\dot{q}_{t+1}\| \leq \bar{v}$ and $\|\ddot{q}_t\| \leq \bar{a}$ uniformly. Define $\psi_t = \dot{h}_t + k_1 h_t$ and*

$$\begin{aligned} C_h &= L_1 \bar{a} + \frac{1}{2} L_2 \bar{v}^2, \\ C_\psi &= 2L_2 \bar{a} \bar{v} + L_2 \bar{a}^2 \Delta t + \frac{1}{2} L_3 \bar{v}^3 + k_1 C_h. \end{aligned}$$

Then

$$\psi_{t+1} \geq (1 - k_2 \Delta t) \psi_t - C_\psi \Delta t^2, \quad (30)$$

$$h_{t+1} \geq (1 - k_1 \Delta t) h_t + \Delta t \psi_t - C_h \Delta t^2. \quad (31)$$

Consequently, if initially $\psi_0 \geq -C_\psi \Delta t / k_2$ and $h_0 \geq -\underline{h}$, where

$$\underline{h} = \frac{\Delta t}{k_1} \left(C_h + \frac{C_\psi}{k_2} \right), \quad (32)$$

then $h_t \geq -\underline{h}$ at every rollout step.

Proof. Write $d_t = q_{t+1} - q_t = \Delta t \dot{q}_{t+1}$. Taylor expansion gives $h_{t+1} = h_t + \Delta t \dot{h}_t + E_h$, where

$$E_h = \Delta t^2 \nabla h(q_t)^\top \ddot{q}_t + \frac{1}{2} d_t^\top \nabla^2 h(\xi_t) d_t. \quad (33)$$

The stated bounds give $|E_h| \leq C_h \Delta t^2$. Expand $\nabla h(q_{t+1})^\top \dot{q}_{t+1}$ around q_t . The first-order term is $\Delta t \dot{h}_t$; the remainder obeys

$$|E_v| \leq (2L_2 \bar{a} \bar{v} + L_2 \bar{a}^2 \Delta t + \frac{1}{2} L_3 \bar{v}^3) \Delta t^2. \quad (34)$$

Combine this expansion with $\ddot{h}_t + k_1 \dot{h}_t \geq -k_2 \psi_t$ to obtain (30); (31) follows from the expansion of h . The lower limits in the statement are fixed points of these two monotone recursions, proving invariance. $\square$

Applying the lemma to $\bar{h} = h - h_{\text{safe}}$ with $h_{\text{safe}} \geq \underline{h}$ keeps the original rollout barrier nonnegative, provided the shifted initial-layer conditions hold. For fixed continuous-time gains the layer is $O(\Delta t)$; with fixed admissible per-step gains $k_i \Delta t$, it is $O(\Delta t^2)$ under the uniform derivative and motion bounds. This is a rollout-model result. An executed system with different integration or dynamics requires a separate discrepancy or sampled-data argument.

For intuition, steady sliding along a boundary of radius ρ at tangential speed v gives a leading adverse order-1 layer $-v^2 \Delta t / (2\rho\gamma)$ on the concave side of the signed-distance barrier. The semi-implicit order-2 layer on the opposite curvature is approximately $-v^2 \Delta t (1/k_1 + 1/k_2) / (2\rho)$. These are asymptotic estimates under steady sliding, not replacements for (32).

B. Virtual-equality error dynamics

Let $e = c_v(q)$ and impose (3) without relaxation. For scalar gains, the linear part of the semi-implicit error update is

$$\begin{bmatrix} e_{t+1} \\ \dot{e}_{t+1} \end{bmatrix} = A_\Delta \begin{bmatrix} e_t \\ \dot{e}_t \end{bmatrix} + r_t, \quad (35)$$

where

$$A_\Delta = \begin{bmatrix} 1 - K_P \Delta t^2 & \Delta t(1 - K_D \Delta t) \\ -K_P \Delta t & 1 - K_D \Delta t \end{bmatrix}. \quad (36)$$

The characteristic polynomial is

$$z^2 - (2 - K_P \Delta t^2 - K_D \Delta t)z + 1 - K_D \Delta t. \quad (37)$$

The Jury conditions reduce to

$$0 < K_D \Delta t < 2, \quad 0 < K_P \Delta t^2 < 4 - 2K_D \Delta t. \quad (38)$$

With $K_P \Delta t^2 = K_D \Delta t = 1$, the linear position terms cancel. Taylor's theorem gives the exact scalar position remainder

$$\begin{aligned} e_{t+1} &= \Delta t^2 \left(\frac{1}{2} \nabla^2 c_v(\xi_t) [\dot{q}_{t+1}, \dot{q}_{t+1}] \right. \\ &\quad \left. - \nabla^2 c_v(q_t) [\dot{q}_t, \dot{q}_t] \right). \end{aligned} \quad (39)$$

Thus $|e_{t+1}| \leq \frac{3}{2} \Delta t^2 \|\nabla^2 c_v\| \bar{v}^2$ if both velocities and the Hessian are bounded. The result applies componentwise to diagonal gains. Under an exact zero-order-hold model, the corresponding deadbeat linear gains instead satisfy $K_P \Delta t^2 = 1$ and $K_D \Delta t = 3/2$. Neither choice removes an unknown execution disturbance; the actual error must be evaluated separately.

C. Executed-only versus per-rollout retraction

Compare two rollout schemes using the same raw perturbations. Scheme A integrates projected inputs without retraction, while scheme B retracts after each step (in phase space at order 2). Assume: (i) retraction displacement is bounded by C_R times equality residual; (ii) bounded projected inputs leave residual at most $C_c \Delta t^2$ after one step from the manifold; (iii) the projected step is Lipschitz in state with constant $1 + L_\Phi \Delta t$; and (iv) the running and terminal costs are Lipschitz with constants L_ℓ, L_φ , with the running cost weighted by Δt .

Discrete Grönwall gives

$$\|x_t^A - x_t^B\| \leq C_R C_c t \Delta t^2 e^{L_\Phi t \Delta t}. \quad (40)$$

Summing the cost differences yields $|S^A - S^B| \leq \delta_S$, where

$$\delta_S = C_R C_c e^{L_\Phi H \Delta t} \Delta t^2 \left(L_\ell \Delta t \frac{H(H+1)}{2} + L_\varphi H \right). \quad (41)$$

For the same fixed temperature β , normalization of the tilted measures implies

$$e^{-2\delta_S/\beta} \leq \frac{dQ^A}{dQ^B} \leq e^{2\delta_S/\beta}. \quad (42)$$

Hence the raw averages satisfy

$$\|\hat{V}^A - \hat{V}^B\| \leq \left(e^{2\delta_S/\beta} - 1 \right) \max_k \|V^k\|. \quad (43)$$

The same-state final motion projection inherits the bound in its metric by nonexpansiveness. First-step projected reactions are identical between the two schemes for matched raw samples

at the measured state; their weighted means differ by at most $(e^{2\delta_S/\beta} - 1) \max_k \|\lambda_0(a_0^k)\|$. Equation (8) combines the two channel bounds for the executed command. At fixed horizon length the state and cost discrepancies are $O(\Delta t^2)$; at fixed horizon time $H\Delta t$ they are $O(\Delta t)$. This conclusion requires the stated uniform boundedness and regularity, as well as matched fixed temperature. Unbounded white-noise scaling, a non-Lipschitz active-set change, or changing cost normalization can invalidate the rate. It is not a claim that omitting retraction is harmless for arbitrary stochastic rollouts.

D. Finite-sample update at fixed temperature

Let V_k be independent draws from a fixed proposal P , $g(V) = e^{-S(V)/\beta}$, $0 < \mathbb{E}_P g < \infty$, and

$$\rho = \frac{\mathbb{E}_P g^2}{(\mathbb{E}_P g)^2} < \infty. \quad (44)$$

For a vector-valued test function z with $|z_i(V)| \leq \bar{z}_i$, define

$$\hat{z}_K = \frac{\sum_k g(V_k) z(V_k)}{\sum_k g(V_k)}, \quad z^* = \mathbb{E}_{Q^*} z. \quad (45)$$

The strong law gives almost-sure convergence. The bounded-test-function self-normalized importance-sampling bound [28], applied coordinatewise, gives

$$\mathbb{E} \|\hat{z}_K - z^*\|^2 \leq \frac{4\rho \|\bar{z}\|^2}{K}. \quad (46)$$

At a fixed cycle, clipped perturbations and a fixed finite nominal sequence bound V , so $z(V) = V$ is admissible. For the first-step projected reaction in a bounded Ω_0 , the same argument applies to $z(V) = \lambda_0(a_0(V))$. Projection of the raw motion mean is nonexpansive in the M -norm. Together with (9), these statements transfer the two channel error bounds to the command obtained by combining their population targets. This is a cycle-wise statement conditional on the nominal sequence, not a uniform bound on all future warm starts.

The targets are tilted means for the specified proposal and cost, not a globally optimal constrained trajectory. Also, the implementation estimates temperature from the same batch of costs. Conditioning on that estimated temperature does not restore the independent fixed-weight setting used in this argument. The numerical controller therefore uses this lemma as a fixed-temperature characterization, while its constraint guarantee follows independently from motion projection and convex reaction averaging.

E. Warm starts and sampling interpretation

For the warm start, the implementation evaluates $\tilde{\lambda}_t = \sum_k w_k \lambda_0(a_t^k)$ using the *measured-state* reaction map at every horizon position. It shifts $\hat{v}_1, \dots, \hat{v}_{H-1}$, appends a zero seed, and replaces their reactions by $\hat{\lambda}_1, \dots, \hat{\lambda}_{H-1}, \hat{\lambda}_{H-1}$ using the correction in (20). The terminal seed therefore has zero tangential input but need not be the zero command. For $t > 0$, these mapped values need not equal the actual rollout reactions or belong to Ω_0 ; they initialize the next search and do not supply its feasibility certificate.

For fixed covariance, the raw and projected averages also have distinct statistical targets. If $Q^* \propto e^{-S/\beta} P$ over raw

sequences, they estimate $\mathbb{E}_{Q^*} V$ and $\mathbb{E}_{Q^*} \Pi(V)$, respectively. These are the means minimizing Gaussian cross-entropy to the tilted raw measure and to its projected pushforward. This is a moment-fitting interpretation; it does not require the projected distribution to be Gaussian.

The implementation scales temperature by the batch cost standard deviation, $\beta = \max(\eta\sigma_S, 10^{-12})$. Feasibility and (17) hold for any resulting nonnegative normalized weights. In contrast, the finite-sample importance-sampling statement in Appendix A-D assumes a fixed temperature independent of the current sample batch. That bound is not claimed for the adaptive-temperature implementation.

F. Experimental protocols and full writing comparisons

Controlled implementation variants: A1 replaces selected support or reaction rows in the rollout with quadratic penalties. A2 uses those penalty rollouts and projects the raw output through an execution filter. A3 is a one-step task-space quadratic program with the same relevant hard rows and a PD tracking objective; it has no predictive horizon. B1 uses a recursive-least-squares point estimate of payload parameters in place of a robust set. These are controlled implementation variants, not reproductions of named external algorithms.

The waiter penalty weights span 10^3 to 10^6 , and A3 gains span 1.5 to 15 rad/s. The best values at $h = 0.14$ m are kept in the height sweep. This is tuning on the main task, not a held-out generalization test. A2 is reported with the same ε as UP-MPPI for the principal equal-margin comparison. The averaging ablation uses raw or projected averages, both with and without that tightening.

Safe-sampling adaptations: Three additional controllers adapt the mechanisms of Shield-MPPI [13], DualGuard MPPI [15], and GS-MPPI [14] to the same robot model, task cost, horizon, and clipped exploration. All retain the common physical-contact dynamics and virtual-equality projection. Their inequality mechanisms differ. The *Shield adaptation* penalizes discrete state-barrier and direct input-row violations, then repairs the first three inputs with at most 20 finite-difference gradient steps. The *DualGuard adaptation* filters samples and the output using individual double-integrator braking sets, a zero-tangential-acceleration support backup, and a reaction-box-center backup. This uses analytic per-joint sets and task-specific backups, rather than a full manipulator reachability computation.

The *GS raw-mean adaptation* composes normalized input-row margins by a soft minimum and applies up to five Euclidean corrections in the virtual-equality null space. It averages raw samples and filters the output at the planning rate. This differs materially from the source GS-MPPI algorithm, which returns the best sampled input for execution and uses a faster inner safety-filter loop [14]. Direct support/reaction rows also enter here at the input-residual level rather than as state barrier functions. Accordingly, these results compare documented local adaptations; they do not reproduce the original algorithms' guarantees or establish a performance ranking of their released implementations.

For the waiter, Shield penalty weights are $C \in \{10^2, 10^3, 10^4\}$ and GS smoothing parameters are $\rho \in$

TABLE VIII

WRITING BASELINES AND LITERATURE-INSPIRED ADAPTATIONS AT ZERO OFFSET (THREE SEEDS). TRACKING IS MEAN $\pm$ SAMPLE STANDARD DEVIATION. METRICS FROM DIVERGENT RUNS COVER ONLY EXECUTED PORTIONS.

Controller	Tracking [mm] $\downarrow$	In band [%] $\uparrow$	Tilt [deg] $\downarrow$	Completed
A1, $w_s = 1$	5.37 ± 0.13	87.5	0.11	3/3
A1, $w_s = 10$	23.08 ± 0.82	87.8	0.12	3/3
A1, $w_s = 100$	54.72 ± 0.62	92.3	0.15	0/3
A1, $w_s = 1000$	67.81 ± 0.78	94.4	0.15	3/3
A2, $w_s = 1$	5.37 ± 0.13	87.5	0.11	3/3
A2, $w_s = 10$	23.08 ± 0.82	88.0	0.12	3/3
A2, $w_s = 100$	54.71 ± 0.63	92.2	0.15	0/3
A2, $w_s = 1000$	67.80 ± 0.78	94.4	0.15	3/3
A3, one-step QP	1.00 ± 0.00	100.0	0.10	3/3
UP-MPPI, GPU	1.58 ± 0.00	99.3	0.11	3/3
DualGuard adaptation	1.55 ± 0.01	100.0	0.11	3/3
GS raw-mean adaptation $\rho = 100$	1.32 ± 0.00	47.0	0.08	3/3
GS raw-mean adaptation $\rho = 20$	1.32 ± 0.00	47.0	0.08	3/3
GS raw-mean adaptation $\rho = 5$	1.32 ± 0.00	50.7	0.08	3/3
Shield adaptation $C = 10$	4.56 ± 0.11	87.4	0.11	3/3
Shield adaptation $C = 100$	15.08 ± 0.61	86.7	0.12	3/3
Shield adaptation $C = 1000$	50.00 ± 0.60	88.3	0.14	0/3

A1 penalizes the reaction band. A2 also filters the raw output. UP-MPPI additionally projects reactions within rollouts and averages those projected reactions; the A2 comparison changes both operations. Blue identifies UP-MPPI; bold marks the best displayed tracking, band occupancy, and tilt among completed runs, including ties. Other controllers retain their better values.

$\{5, 20, 100\}$. For each numerical margin, the variant with the lowest mean first-move RMSE and no tips at $h = 0.14$ m is used in the height sweep. Writing evaluates all seven configurations listed in Table VIII, with Shield weights 10, 100, 1000. Dual-arm uses $C = 1000$ and $\rho = 20$; these three variants have neither the acceleration-error tightening nor the reaction box used by UP-MPPI.

Optimization comparators: An offline robust plan follows the plan-and-track structure of Heins and Schoellig [22], adapted here to joint-acceleration inputs, 50 ms knots, IPOPT, robust support rows with quadratic slacks, and acceleration-level PD tracking. It knows the entire reference in advance. Online robust NMPC uses the MPPI horizon, step, and tracking-cost terms, plus jerk regularization; support rows are hard at the first knot and softened with quadratic slacks at future knots. The lowest-error tested jerk weight at the main height is $w_j = 0.1$. Each optimizer is deterministic and is evaluated once per condition. Different discretizations, slack treatment, and additional regularization preclude attributing all performance differences to the optimization family.

Full writing sweep: The writing comparison tests reaction-penalty weights $w_s \in \{1, 10, 100, 1000\}$ at three offsets and three seeds, together with A3 and GPU UP-MPPI. Table VIII reports every tested weight at zero offset, including divergent runs. A2 filters its raw output but does not average clamped rollout reactions; its comparison with UP-MPPI changes both in-rollout reaction handling and reaction averaging. The motion-only ablation in Section IV-C isolates support-projection placement without that difference.

The one-step QP achieves 1.00 mm RMSE and 100.0% force-band occupancy, compared with 1.58 mm and 99.3% for GPU UP-MPPI. DualGuard’s adaptation reaches 100% occupancy at 1.55 mm. GS raw-mean adaptations reach 47–51% occupancy at approximately 1.32 mm. These contact-task results describe the implemented extensions and do not establish that published GS-MPPI cannot handle writing or force control. In particular,

filter construction, averaging, and execution rules differ, so the cross-method differences do not isolate the averaging mechanism.

Transport calibration and measurement bounds: Wrist force and torque measurements have independent uniform per-sample noise bounded by 0.1 N and 3 mNm. Measurement windows are 100 ms, after an initial settling period. Separate oracle runs at $h \in \{0.05, 0.115, 0.18\}$ m and separate seeds set each error bound to 1.5 times its largest observed calibration residual. The resulting force and torque bounds are 0.033 N and 3.5 mNm. The direct estimator allows 2.5 mm object-offset uncertainty; the tracked variant uses 0.5 mm. COP differences use a creep bound of 0.1 mm plus 1.7 mm/s times the window separation and duration.

These are empirical calibration assumptions, not distribution-free worst-case bounds. The force-window bound is smaller than the per-sample noise bound because of averaging. Evaluation therefore checks parameter containment and actual residual-to-bound ratios. Updates are gated by a period-averaged COP within 90% of the support edge. This is a sticking proxy and does not independently certify absence of slip.

The execution tightening is $\varepsilon = 0.165 \text{ m/s}^2$, from a largest calibration deviation of 0.110 m/s^2 . It applies to the period-averaged discrepancy in s , as in (23). The geometric octagon additionally lies inside the square footprint; this geometric reserve is distinct from the calibrated execution margin.

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